Fetal Distress Detection from CTG Signals

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Janitri Interview Assignment

1. How I Framed It

I framed this as a binary classification problem: given the CTG signals recorded during labour, predict whether the newborn will be distressed at delivery.

Label definition

A recording is labelled distressed (positive class, y=1) when either:
- Umbilical-cord pH < 7.20, or
- 5-minute Apgar score < 7.

Rationale: pH < 7.20 is a widely-used threshold for clinically significant metabolic acidosis (ACOG, FIGO guidelines). A 5-min Apgar below 7 indicates the baby needed resuscitation and may have been compromised. Using the OR rule captures both biochemical and clinical indicators of distress and produces a larger positive class (~33% of recordings), which helps the model learn meaningful patterns from a modest 552-recording dataset.

Final dataset: 552 recordings total, 370 normal (67%), 182 distressed (33%).

Input representation

Rather than feeding raw 4 Hz signals directly into a model (which would require much more data), I extract 29 hand-crafted time-domain features from the last 30 minutes of each recording. These features capture:
- FHR statistics (mean, std, median, range, IQR, skewness, kurtosis)
- Heart-rate variability (RMSSD, SDNN, short/long-term variability)
- Clinical events (count of decelerations and accelerations)
- Uterine contraction patterns (frequency, amplitude, interval regularity)
- FHR-UC coupling (correlation, FHR during vs. between contractions)
- Signal quality (fraction of missing/artefact FHR values)

These features are motivated by clinical CTG interpretation guidelines (FIGO) and prior literature on computerised CTG analysis.

2. What I Built and How I Checked It

Data processing

I loaded all 552 recordings from the CTU-CHB database using the wfdb library. For each recording I:
1. Read the physical (scaled) FHR and UC signals from the .dat file.
2. Cleaned the FHR signal by replacing out-of-range values (<50 or >250 bpm) and zeros with NaN.
3. Extracted the last 30 minutes of data (7200 samples at 4 Hz).
4. Computed the 29-feature vector.
5. Parsed the .hea header to extract pH, BDecf, Apgar1, Apgar5.

All 552 records were successfully processed. The data was split 80/20 into train (441 samples) and test (111 samples) sets, stratified by label.

Model choice

I chose a Random Forest classifier (scikit-learn) with 300 trees and max depth 12. Key reasons:

- Works well on small tabular datasets without extensive hyperparameter tuning.
- Handles mixed-scale features natively.
- Provides feature importances, which are valuable for clinical interpretability.
- class_weight='balanced' up-weights the minority (distressed) class.

Features are standardised with StandardScaler before training.

Evaluation metrics

I evaluated the model on the held-out 20% test set:

Accuracy: 0.7027
ROC-AUC: 0.7330
Precision: 0.5909
Recall: 0.3514
F1 Score: 0.4407
Specificity: 0.8784

Confusion matrix: TP=13, FP=9, FN=24, TN=65

Metric interpretation:
- ROC-AUC of 0.73 indicates moderate discrimination ability, better than random.
- High specificity (0.88) means the model correctly identifies most healthy cases.
- Low recall (0.35) means the model misses many distressed cases. This is the main weakness and must be addressed for clinical use.
- Precision of 0.59 means about 6 in 10 predicted-positive cases are truly distressed, which is acceptable for a screening tool.

What the metrics do NOT tell us: ROC-AUC does not capture the clinical cost of different error types. In practice, missing a distressed baby (false negative) is far more dangerous than a false alarm (false positive). The threshold should be lowered to improve recall at the cost of some precision.

3. Clinical Utility and Inference

Intended use

This system is imagined as a decision-support tool that runs alongside the CTG monitor in a labour ward. It would NOT replace the clinician; instead it would provide an automated second opinion.

Usage scenario:
1. The CTG monitor streams FHR and UC data continuously.
2. Every N minutes (e.g. every 10 min), the system extracts the last 30 minutes of signal and computes the 29-feature vector.
3. The model produces a distress probability (0-1) and a binary alert.
4. If the probability exceeds a configurable threshold, a flag appears on the bedside display, prompting the nurse or doctor to review the trace.

The threshold can be tuned to the clinical context: a lower threshold catches more true positives but raises more false alarms; a higher threshold reduces alarms but risks missing some cases.

generate_outcomes.py

The inference script (generate_outcomes.py) implements the prediction pipeline:
- Input: either a raw WFDB record path or a pre-extracted .npz feature file.
- Processing: loads model.pkl and scaler.pkl from artifacts/, extracts features (if given a raw record), scales them, and runs the Random Forest.
- Output: for each input, prints the distress probability (float, 0-1) and the binary prediction (0=not distressed, 1=distressed).

This mirrors the real-world deployment: the model receives a window of CTG data, computes features, and returns a probability. The clinical team acts on the alert.

4. Limits and Next Steps

Known limitations

1. Small dataset (552 recordings): limits model complexity and generalisability. Results may not transfer to other hospitals or populations without re-training.

2. Retrospective labels: pH and Apgar are measured at delivery, but the model predicts from a 30-min window that may be well before delivery. The model is learning associations, not causal mechanisms.

3. Low recall (0.35): the model misses a majority of distressed cases. This is the most critical limitation for clinical safety. Threshold tuning, SMOTE, or a different model architecture could improve this.

4. No temporal modelling: the hand-crafted features summarise the window into a single vector, discarding the time-series structure. A sequential model (LSTM, Transformer) could capture progressive deterioration.

5. Signal quality: some recordings have significant artefact (missing FHR). The current approach treats missing values as NaN and computes features from the valid portion, but this could bias estimates.

6. No external validation: the model has only been tested on a random split of the same dataset. A proper external validation on a different hospital's data is essential before any clinical use.

Next steps with more time

- Hyperparameter tuning with cross-validation (grid search or Bayesian).
- Try gradient-boosted trees (XGBoost, LightGBM) which often outperform RF.
- Experiment with a 1-D CNN or LSTM on the raw 4 Hz signal.
- Sliding-window inference: produce a probability every 5-10 minutes to track how the risk evolves during labour (a risk trajectory).
- Calibration: use Platt scaling or isotonic regression so the output probability better reflects the true prevalence.
- SHAP values for per-prediction explanations.
- External validation on a second dataset.